

The Geometry of Adversarial Identity Cost: Structural Limits of Sybil Resistance in Open Systems

Anonymous Author(s)

Abstract—Open and permissionless systems mitigate Sybil attacks by binding influence to scarce resources. However, *aggregate scarcity alone does not determine how the cost of sustaining multiple identities scales over time*. Many widely deployed resources—such as computation and financial capital—are *divisible and reusable*: they can be subdivided, transferred, and reapplied across identities and time without proportional marginal expenditure. As a result, adversarial effort may be amortized across identities, enabling large-scale identity replication at sublinear cost.

This paper develops an axiomatic framework for analyzing security-weighting resources through the lens of *adversarial cost geometry*. We identify a small set of resource properties—including divisibility, additivity of influence, temporal reusability, identity transferability, and throughput boundedness—that characterize the asymptotic growth of sustained adversarial cost.

Under a standard amortized coordination model, we establish an asymptotic boundary between two resource classes. Within the present abstraction, mechanisms grounded in divisible, reusable resources admit identity amortization, yielding $C(s, T) = o(sT)$; the marginal cost per additional identity does not grow with the time horizon. In contrast, throughput-bounded, non-transferable, window-local resources indicate a linear scaling regime, $C(s, T) = \Omega(sT)$, with marginal cost $\Delta(s, T) = \Omega(T)$.

Under the assumptions considered here, protocol rules that redistribute a fully divisible and reusable resource may be insufficient to enforce linear per-identity cost. The framework is intended as a resource-centric asymptotic lens that complements existing deployment-level and protocol-level analyses rather than replacing them. We accompany the theoretical results with analytical illustrations of the scaling regimes across varying identity counts, time horizons, and coordination overhead models.

Index Terms—Sybil attacks, adversarial cost models, distributed security, resource parallelizability, identity scaling.

I. INTRODUCTION

A. Motivation

Open and permissionless systems must regulate influence under adversarial conditions in which identity creation is inexpensive and effectively unbounded. The classical formulation of this problem is the Sybil attack [1], where a single adversarial entity generates multiple identities to amplify control over a distributed system. This vulnerability affects peer-to-peer overlays, blockchain consensus [2]–[6], decentralized governance, reputation systems, and distributed access-control frameworks [7]–[10].

A dominant design paradigm mitigates Sybil attacks by binding influence to the possession or expenditure of a scarce resource. Proof-of-Work (PoW) associates block-production probability with computational effort [5], while Proof-of-Stake

(PoS) binds influence to locked financial capital [11]. Such mechanisms have been analyzed under partially synchronous network models, yielding safety and liveness guarantees in terms of aggregate adversarial resource ownership [5], [11], [12].

However, many security-critical operations in open systems function at the *identity layer* rather than the resource layer. Examples include decentralized governance [22], committee selection [23], reputation systems [24], rate-limited participation [25], and identity-based access control. In these settings, the relevant security question shifts from how much resource an adversary controls in aggregate to how the cost of *sustaining multiple identities* scales with their number and the duration of participation.

A Resource-Structural Perspective. A frequently implicit property of widely deployed resources is that they are *divisible and reusable*: computational capacity and financial capital can be subdivided, delegated, and reapplied across identities without proportional marginal expenditure. Mining pools in PoW and stake delegation in PoS illustrate this—once aggregate resource ownership is acquired, it can be redistributed across arbitrarily many identities while preserving total influence [9], [11], [26], [27]. As a result, adversarial effort may be *amortized across identities*: a single adversary can sustain many identities while incurring only slowly growing marginal cost.

This observation reveals an asymptotic limitation of scarcity-based defenses not captured by existing analyses. Economic models focus on equilibrium cost-of-corruption [29], [30], while protocol-level analyses focus on aggregate adversarial influence [31], [32]. Neither framework isolates the resource properties that govern how adversarial cost scales with the number of sustained identities.

This motivates the following question:

What properties must a security resource possess for sustained adversarial cost to grow proportionally with the number of active identities?

B. Contribution and Framing

This paper develops a resource-centric asymptotic lens for analyzing identity-level Sybil resistance. The objective is not to replace deployment-specific or protocol-level security analyses, but to isolate how the structural properties of a security-weighting resource determine the asymptotic geometry of adversarial identity cost. The framework is intended to complement existing analyses—including aggregate-resource

consensus models and economic equilibrium analyses—by identifying when divisibility and reusability enable adversarial cost amortization.

Under a standard amortized coordination model, we establish an asymptotic boundary between two resource classes:

- Resources satisfying *divisibility*, *additivity*, *transferability*, and *temporal reuse* admit identity amortization within the considered model, yielding $C(s, T) = o(sT)$.
- Resources that are *rate-limited*, non-transferable, and window-local indicate the scaling regime $C(s, T) = \Omega(sT)$.

Within the considered model, this boundary reflects the resource class of the underlying mechanism rather than its protocol-level design. Fully divisible and reusable resources admit sublinear per-identity cost regardless of the protocol rules designed around them, within the present abstraction. Rate-limited, non-transferable resources, by contrast, impose per-identity participation constraints that resist amortization.

As a secondary implication, divisibility and reusability induce *centralization pressure*: because resource aggregation does not incur proportional identity-level cost, participants are incentivized to pool or delegate, consistent with empirically observed behavior in mining pools and delegation markets [7], [28].

C. Contributions

- **Axiomatic Resource Framework.** We introduce a framework based on properties including divisibility, additivity of influence, temporal reusability, identity transferability, and throughput boundedness.
- **Asymptotic Limit for Divisible Resources.** We establish that mechanisms grounded in fully divisible, reusable resources admit identity amortization within the considered model, yielding $C(s, T) = o(sT)$.
- **Linear Lower Bound for Rate-Limited Resources.** We establish that throughput-bounded, non-transferable resources imply $C(s, T) = \Omega(sT)$.
- **Asymptotic Scaling Separation.** We establish that the two resource classes induce asymptotically disjoint cost regimes under the boundary conditions characterized in this work.
- **Design Implications.** Under the considered model, identity-level cost scaling is constrained by resource structure rather than protocol syntax.

D. Organization of the Paper

Section II reviews related work. Section III introduces the system model. Section IV formalizes the axiomatic framework. Sections V and VI establish the main bounds. Section VII proves the scaling separation theorem. Section VIII presents concrete resource instantiations. Sections IX and X discuss implications and limitations. Section XI presents analytical illustrations of the scaling regimes.

II. RELATED WORK

Prior research has addressed adversarial influence and Sybil attacks in open systems from several perspectives. However,

these research directions typically study security questions that differ from the one considered in this paper. In particular, existing work largely focuses on identity admission, consensus safety, or economic incentives, while the present work studies how the *structural properties of security resources determine the asymptotic cost of sustaining multiple identities over time*.

Table I summarizes the distinction between major research directions and the focus of this paper.

A. Resource-Based Sybil Defenses

The Sybil attack, formally introduced by Douceur [1], established that identity replication in open systems cannot be prevented without binding influence to scarce resources. Modern permissionless systems implement this principle through resource-weighted mechanisms.

Nakamoto’s proposal [46] introduced Proof-of-Work (PoW), in which block-production probability scales with computational effort. Subsequent formal analyses [5], [6], [38], [39] established security properties of Nakamoto-style consensus, including chain growth, common prefix, and chain quality guarantees.

Proof-of-Stake (PoS) systems replace computational effort with locked financial capital. Protocols such as Ouroboros [11] and its extensions [47], [48] provide provable security guarantees under partially synchronous network models, while protocols such as Algorand [23], [49] use cryptographic sortition to sample committees proportionally to stake.

Empirical and systems-level analyses further study the security and performance of blockchain consensus mechanisms. Work on mining incentives, pooling behavior, and adversarial strategies [13], [14], [42], [43] demonstrates how adversarial resources are frequently aggregated, pooled, and redistributed across multiple logical identities. Other studies analyze decentralization and deployment realities [7], [9], [50], [51].

Additional work studies network-level adversaries such as eclipse and routing attacks [18], [19], [52], as well as economic stability of mining rewards and fee mechanisms [15], [29], [30], [41].

Despite this extensive literature, these analyses consistently reason about *aggregate adversarial resource share*. Security guarantees are expressed in terms of the fraction of computational power or stake controlled by an adversary. However, these models do not explicitly analyze how adversarial cost scales with the number of sustained identities.

B. Graph-Based and Social-Structure Defenses

Another line of work constrains Sybil proliferation using the structure of social networks. Mechanisms such as SybilGuard [33] and SybilLimit [34] exploit the limited number of attack edges between honest and adversarial regions of a social graph to restrict the influence of Sybil identities.

Later work such as SybilInfer [35] applies probabilistic inference over trust graphs to detect clusters of suspicious nodes. These approaches can significantly increase the difficulty of creating large Sybil regions when the social graph exhibits strong connectivity properties.

TABLE I
POSITIONING OF THIS WORK RELATIVE TO PRIOR LITERATURE

Research Line	Primary Security Question	Representative Works
Sybil Attack Literature	Can adversaries create multiple identities, and how can identity creation be restricted or detected?	[1], [33]–[37]
Blockchain Consensus Security	Under what conditions does aggregate adversarial resource ownership violate safety or liveness properties of consensus protocols?	[5], [6], [11], [38]–[40]
Economic Analyses of Consensus	What equilibrium incentives and corruption costs arise in resource-weighted consensus systems?	[29], [30], [41]–[43]
Identity-Binding Mechanisms	How can systems enforce one-person–one-identity mappings through ceremonies or identity verification?	[44], [45]
This Paper	How do structural properties of security resources determine the asymptotic cost of sustaining multiple identities over time?	Structural analysis of identity-scaling cost

However, their guarantees depend heavily on the topology and trust assumptions of the underlying network. When such assumptions fail, these mechanisms do not provide explicit bounds on how adversarial cost scales with the number of sustained identities. More generally, these approaches constrain the *placement* of identities within a social graph, but do not characterize the *cost of sustaining identities* as their number grows.

C. Randomness and Committee Sampling

Leader election and committee sampling mechanisms often rely on cryptographic randomness primitives. Verifiable random functions (VRFs) [53] enable unpredictable leader selection, while bias-resistant randomness protocols [54] enable robust distributed randomness generation.

Composable analyses of blockchain protocols [5], [38], [55] demonstrate robustness under dynamic participation and partially synchronous networks. More recent asynchronous constructions such as HoneyBadgerBFT [31] illustrate that scalable consensus can be achieved without strong synchrony assumptions.

Nevertheless, these mechanisms continue to model adversarial capability in terms of aggregate resource ownership. Randomness primitives improve scalability and fairness of leader selection but do not alter the structural divisibility or transferability of the underlying resource.

D. Human-Centric Identity Mechanisms

Human-centric approaches attempt to enforce one-person–one-identity mappings through physical ceremonies, biometric verification, or behavioral signals. Proof-of-Personhood protocols [44] represent one prominent example, while later work explores pseudonym parties and social identity verification mechanisms [45].

Such systems aim to restrict identity creation by binding credentials to real-world participants. However, these mechanisms primarily address identity admission rather than sustained participation. Once identities are admitted, influence may still be governed by reusable or transferable resources.

Consequently, identity admission mechanisms alone do not guarantee linear adversarial cost over time.

E. Rate Limiting and Capability Systems

Systems research has long studied per-actor resource constraints. Capability-based security architectures [56] limit authority through unforgeable tokens, while congestion-control algorithms [57] regulate per-flow network throughput.

More recent hardware-based security architectures, including trusted execution environments such as Intel SGX [58], enable systems to bind computation to specific devices or execution contexts, enforcing hardware-rooted execution constraints.

Such mechanisms impose intrinsic limits on per-identity or per-device throughput, effectively restricting the rate at which independent participants can generate verifiable activity.

Although these frameworks effectively enforce per-actor or per-device rate limits, they are typically analyzed in terms of access control, network stability, or trusted hardware guarantees rather than adversarial identity scaling.

F. Positioning of This Work

This paper departs from prior research in two key ways. First, rather than analyzing specific protocols or admission mechanisms, we develop an axiomatic theory of *security-weighting resources*. We formalize structural properties such as divisibility, additivity of influence, temporal reusability, identity transferability, and throughput boundedness, and analyze how these properties determine the geometry of adversarial cost. Second, we establish a structural separation at the extremal boundary conditions between resource classes. Parallelizable resources induce sublinear sustained identity cost within the present model, whereas throughput-bounded, non-transferable, window-local resources enforce

$$C(s, T) = \Omega(sT).$$

Unlike prior work on cost-of-corruption or equilibrium security [29], [30], our framework does not analyze the economic

TABLE II
CORE NOTATION

Symbol	Meaning
v	Identity (participant instance)
t	Time window index
T	Time horizon (number of windows)
s	Number of sustained identities
$r_v(t)$	Resource allocated to identity v at time t
$W_v(t)$	Influence of identity v at time t
$f(\cdot)$	Resource-to-influence mapping
S_t	Set of adversarial identities at time t
$C(s, T)$	Cost to sustain s identities over T windows
$\Delta(s, T)$	Marginal identity cost

threshold required to corrupt a protocol instance. Instead, it studies how the structural properties of the underlying resource determine the asymptotic cost of sustaining multiple identities over time. Existing economic analyses characterize aggregate adversarial expenditure under equilibrium assumptions, while Sybil-defense mechanisms [25] typically focus on identity admission or detection. In contrast, the present work isolates structural conditions under which adversarial identity cost becomes amortizable or remains proportional to the number of sustained identities—a dimension that neither aggregate cost models nor admission-based defenses address. Thus, unlike prior work that focuses on identity creation, consensus thresholds, or economic equilibria, our analysis identifies identity-level Sybil resistance as a consequence of the structural properties of the underlying resource class. Specifically, prior work does not isolate whether a fixed aggregate resource can be decomposed across many identities at sublinear marginal cost—the central question of this paper.

III. SYSTEM MODEL

We consider an open and permissionless system in which identities are inexpensive to create and no trusted admission authority exists. Participation is therefore identity-unbounded, and adversaries may generate arbitrarily many identities at negligible syntactic cost.

Security mechanisms regulate influence by binding participation to the possession or expenditure of a resource R . Our objective is to characterize how structural properties of R determine the scaling behavior of adversarial identity replication.

A. Notation

Table II summarizes the core notation used throughout the paper.

B. Open Identity Model

Let $\mathcal{U}$ denote the universe of potential participants. At each time window t , the system contains a set of active identities V_t .

An adversary $\mathcal{A}$ is identity-unbounded: it may create, retire, and coordinate an arbitrary number of identities at negligible syntactic cost, consistent with the Sybil model [1].

Let $S_t \subseteq V_t$ denote the subset of identities controlled by $\mathcal{A}$ during window t , with $|S_t| = s_t$. We impose no a priori bound on s_t .

An identity is an abstract participant instance capable of receiving resource allocation and producing influence.

C. Time Model

Time is divided into discrete windows

$$t = 1, 2, \dots$$

A window represents the minimal interval over which resource allocation and influence may be refreshed (e.g., block intervals or governance epochs).

We evaluate security over a finite horizon of T windows. An identity is said to be *sustained* if it remains active throughout all windows $t = 1, \dots, T$.

D. Resource and Influence

Let R denote the security-weighting resource. For each identity $v \in V_t$ and window t ,

$$r_v(t) \in \mathbb{R}_{\geq 0}$$

denotes the allocated resource. Influence is determined by a mapping

$$W_v(t) = f(r_v(t)).$$

We do not impose structural assumptions on f at this stage; these will be derived from properties of the resource R .

Definition 1 (Activation Threshold). *There exists a constant $r_{\min} > 0$ such that an identity is active in window t only if*

$$r_v(t) \geq r_{\min}.$$

This captures the requirement that participation requires a strictly positive resource commitment.

Remark 1. *The activation threshold $r_{\min}$ establishes a per-identity floor that any active identity must meet, regardless of resource structure. This baseline cost is unavoidable and grows linearly with the number of sustained identities. The structural separation established in Sections V–VII does not concern this baseline. Rather, it asks whether a resource imposes an additional per-identity, per-window cost beyond $r_{\min}$. Specifically, Theorem 1 shows that a fully parallelizable resource may not impose such additional cost within this model: an adversary need only acquire an aggregate allocation $r \geq s r_{\min}$ and redistribute it across s identities, incurring no further T -scaling overhead. Theorem 2 shows that a throughput-bounded resource does impose this additional structure, making $r_{\min} \cdot T$ the unavoidable marginal cost per new identity.*

The total adversarial influence at time t is

$$W_{\mathcal{A}}(t) = \sum_{v \in S_t} W_v(t).$$

E. Adversarial Model

The adversary $\mathcal{A}$ is computationally unbounded but constrained by the structural properties of the resource R .

Specifically, $\mathcal{A}$ may:

- Instantiate arbitrarily many identities.
- Allocate and redistribute resource across identities when permitted by the resource structure.
- Reuse or reallocate resource across time windows when temporal reuse is allowed.
- Coordinate identities without communication constraints.

This model subsumes aggregate-resource adversaries used in blockchain analyses [5], [11], while remaining protocol-independent.

F. Model Scope and Coordination Assumptions

Our adversarial model assumes that identity coordination does not itself impose throughput constraints comparable to those enforced by the underlying security resource.

In particular, we distinguish between: (i) resource-induced constraints, which arise from the structural properties of the security-weighting resource, and (ii) coordination overhead, which captures auxiliary costs associated with managing multiple identities.

Throughout the paper, coordination costs are modeled separately through the function $h(s, T)$, allowing the structural properties of the resource to be analyzed independently of implementation-specific orchestration mechanisms.

The results established in Sections V–VII therefore isolate the scaling implications induced by resource structure rather than by assumptions about communication complexity, software architecture, or operational deployment details.

G. Adversarial Cost

We define the adversarial cost function $C(s, T)$ as the minimum total resource expenditure required to sustain at least s active identities over T consecutive windows.

To make the structure of costs precise, we decompose $C(s, T)$ into three components:

$$C(s, T) = C_{\text{stock}}(s) + C_{\text{flow}}(s, T) + h(s, T),$$

where:

- $C_{\text{stock}}(s)$ is the one-time acquisition cost of resources that persist across all T windows without renewal (e.g., staked capital, hardware);
- $C_{\text{flow}}(s, T)$ is the cumulative per-window expenditure for resources that must be renewed each window (e.g., energy, device attestations, human participation);
- $h(s, T)$ is the coordination overhead of managing s identities over T windows.

Formally,

$$C(s, T) = \inf \left\{ \sum_{t=1}^T \mathcal{R}_{\mathcal{A}}(t) \mid |S_t| \geq s \ \forall t \in \{1, \dots, T\} \right\}.$$

The structural separation of Sections V–VII concerns the scaling of $C_{\text{flow}}(s, T)$: parallelizable resources admit $C_{\text{flow}}(s, T) = 0$ (stock suffices for all windows), whereas throughput-bounded resources enforce $C_{\text{flow}}(s, T) = \Omega(sT)$. The coordination term $h(s, T)$ is governed by Assumption 1.

H. Scaling Regimes

We distinguish two asymptotic regimes:

• Sublinear Scaling:

$$C(s, T) = o(sT)$$

• Linear Scaling:

$$C(s, T) = \Omega(sT)$$

Linear scaling implies that each additional sustained identity incurs proportional per-window cost. Sublinear scaling implies identity amortization, where marginal identity cost vanishes asymptotically.

I. Objective

Our goal is to characterize how structural properties of the resource R determine which scaling regime arises.

In particular, we show that adversarial identity scaling is shaped not by protocol design alone, but by intrinsic structural constraints of the underlying resource.

IV. AXIOMATIC THEORY OF SECURITY RESOURCES

We develop an axiomatic framework for security-weighting resources that isolates the structural properties responsible for adversarial identity scaling.

Rather than analyzing specific protocols, we model the underlying resource R used to assign influence to identities. Our goal is to characterize how intrinsic structural properties of R determine whether identity amplification is possible.

Throughout the analysis, let $r_v(t) \in \mathbb{R}_{\geq 0}$ denote the resource allocation assigned to identity v during window t , and let $W_v(t) = f(r_v(t))$ denote the resulting influence.

A. Resource as a Structural Object

We model a security-weighting resource as a quantity that

- 1) can be allocated across identities,
- 2) evolves across time windows, and
- 3) determines influence through an allocation rule f .

Our analysis is agnostic to the semantic nature of the resource (e.g., computation, capital, or attention). Instead, it depends only on structural properties governing division, aggregation, reuse, and transfer across identities and time.

B. Structural Properties

Definition 2 (Divisibility). *A resource R is divisible if for any allocation $r \geq 0$ and integer $k \geq 1$ there exist allocations $r_1, \dots, r_k \geq 0$ such that*

$$\sum_{i=1}^k r_i = r,$$

and the adversary may distribute r_i across k identities without additional structural cost.

Definition 3 (Additivity of Influence). *A resource induces additive influence if*

$$\sum_{v \in S_t} f(r_v(t)) = f\left(\sum_{v \in S_t} r_v(t)\right).$$

Definition 4 (Temporal Reusability). A resource is temporally reusable if allocation in window t may be reused in window $t + 1$ without renewed identity-bound expenditure.

Definition 5 (Identity Transferability). A resource is identity-transferable if an allocation $r_v(t)$ may be reassigned to another identity v' without proportional cost.

Definition 6 (Throughput-Bounded Resource). A resource R is throughput-bounded if participation is mediated through independent execution channels, each bound to a specific identity such that resource cannot be reassigned across identities without proportional cost.

Formally, each identity v is associated with a dedicated channel C_v satisfying all three of the following conditions:

- 1) **Per-channel rate limit:** there exists a constant $\tau > 0$ such that

$$r_v(t) \leq \tau$$

for all v and t ;

- 2) **Non-transferability:** the allocation $r_v(t)$ assigned to identity v cannot be simultaneously credited to any other identity $v' \neq v$;
- 3) **Window-locality:** allocation in window t does not carry over to window $t + 1$; sustained participation therefore requires renewed expenditure in every window.

Parameter relationship. The throughput bound τ and the activation threshold $r_{\min}$ (Definition 1) are independent parameters governing distinct structural constraints. The activation threshold specifies the minimum resource required for an identity to be active in a given window, while τ bounds the maximum resource any single identity can contribute per window. For the system to admit active identities, these parameters must satisfy $0 < r_{\min} \leq \tau$; otherwise no identity can meet the activation threshold within its channel capacity. The linear lower bound of Theorem 2 holds for any such configuration and is tightest when $r_{\min} = \tau$.

Throughput boundedness imposes a per-identity rate constraint that is structurally non-transferable and window-local. Scaling the number of identities therefore requires provisioning additional independent channels.

C. Structural Implications

The above properties jointly determine adversarial scaling behavior.

Proposition 1 (Identity Partition Invariance Under Parallelizable Resources). If a resource R satisfies divisibility, additivity, temporal reusability, and identity transferability, then adversarial influence depends only on aggregate resource ownership and is independent of identity partitioning.

Proof. Let R be a resource satisfying all four properties, and let A be an adversary controlling an aggregate allocation $r = \sum_{v \in S_t} r_v(t)$ across a set S_t of identities at time t .

Step 1: Partitioning does not change aggregate influence. By divisibility (Definition 2), the adversary may partition r

into any collection $r_1, \dots, r_s \geq 0$ satisfying $\sum_{v=1}^s r_v = r$. By additivity of influence (Definition 3),

$$\sum_{v \in S_t} f(r_v(t)) = f\left(\sum_{v \in S_t} r_v(t)\right) = f(r).$$

Hence total influence $W_A(t) = f(r)$ is invariant to how r is partitioned across identities.

Step 2: Reassignment across identities preserves aggregate influence. By identity transferability (Definition 5), the allocation $r_v(t)$ assigned to any identity v may be reassigned to any other identity v' without proportional cost. Combined with Step 1, any redistribution of r across identities leaves $W_A(t)$ unchanged.

Step 3: Persistence across time windows. By temporal reusability (Definition 4), the aggregate allocation r obtained in window t may be reused in all subsequent windows $t+1, \dots, T$ without renewed expenditure. Therefore $W_A(t) = f(r)$ holds uniformly across all windows, independent of identity configuration.

Conclusion. Since $W_A(t) = f(r)$ depends only on the aggregate allocation r and not on how r is distributed across identities or time, adversarial influence is fully determined by aggregate resource ownership. Identity partitioning is therefore structurally irrelevant. $\square$

D. Resource Classes

Definition 7 (Parallelizable Resource). A resource R is parallelizable if it satisfies divisibility, additivity of influence, temporal reusability, and identity transferability.

Definition 8 (Throughput-Bounded Non-Parallelizable Resource). A throughput-bounded non-parallelizable resource is any resource R satisfying Definition 6.

E. Resource Taxonomy

Table III summarizes representative resource classes and the structural properties they satisfy. Proof-of-Work is treated as two distinct components: the reusable *capacity* component (mining hardware) and the non-reusable *operational* component (energy expenditure). The structural results of this paper apply to each component separately; the full economic cost of PoW mining involves both and must be analyzed accordingly.

Parallelizable resources (e.g., staked capital and computational capacity) support subdivision, transfer, and reuse of their stock component, enabling adversarial effort to be amortized across identities and time.

Proof-of-Work is a *hybrid* that must be decomposed for structural analysis. Its *capacity component* (mining hardware) is a reusable stock resource: once acquired, it can be partitioned across identities without renewed expenditure, placing it in the parallelizable class of Theorem 1. Its *operational component* (energy) is non-reusable across windows: it must be renewed each window, contributing a flow cost not captured by the parallelizable classification. Theorem 1 addresses the identity-partitioning behavior of the capacity component; the complete economic cost of mining combines both components and should be analyzed accordingly.

TABLE III
STRUCTURAL TAXONOMY OF SECURITY RESOURCES. PROOF-OF-WORK IS DECOMPOSED INTO ITS CAPACITY COMPONENT (MINING HARDWARE: PARALLELIZABLE STOCK) AND ITS OPERATIONAL COMPONENT (ENERGY: NON-REUSABLE FLOW). THEOREM 1 APPLIES TO THE IDENTITY-PARTITIONING BEHAVIOR OF THE CAPACITY COMPONENT ONLY. THE FULL ECONOMIC COST OF POW INCLUDES BOTH COMPONENTS.

Resource Class	Div.	Add.	Reuse	Transfer	Cost Type	Scaling
PoW: Capacity component (hardware)	✓	✓	✓	✓	Stock	Sublinear
PoW: Operational component (energy)	✓	✓	×	✓	Flow	Sublinear in s , Linear in $T^\dagger$
Capital (PoS)	✓	✓	✓	✓	Stock	Sublinear
Social-Graph Trust	Partial	Partial	Partial	Partial	Mixed	Model-Dependent
Human Real-Time Channel	×	×	×	×	Flow	Linear
Device-Bound Execution	×	×	×	×	Flow	Linear

[†] Energy expenditure is non-reusable across windows (Temporal Reuse = $\times$) but remains transferable across identities and divisible. It therefore produces a flow cost that scales linearly in T (each window requires fresh expenditure) but sublinearly in s (energy can be redistributed across identities). This component falls outside the purely parallelizable class of Theorem 1, which applies only to the capacity (stock) component of PoW.

This decomposition resolves the apparent tension between Theorem 1 and the empirically observed flow costs of PoW: the theorem does not claim that PoW mining is free across time, but that the *capacity* component can be partitioned across identities without additional structural cost.

In contrast, throughput-bounded resources rely on independent execution channels whose capacity cannot be aggregated. Sustaining additional identities therefore requires provisioning additional channels, leading to linear cost scaling.

F. Coordination Cost Assumption

Assumption 1 (Amortized Coordination). *The adversarial cost of coordinating s identities over T windows satisfies*

$$h(s, T) = o(sT).$$

Scope. This is an environmental and deployment assumption, not a structural property of the security-weighting resource R . It models the operational overhead of managing multiple identities—such as software orchestration, account administration, or communication—rather than the resource-induced participation cost captured by $C_{\text{flow}}(s, T)$. The two components are independent: $h(s, T)$ can be reduced through automation or shared infrastructure, whereas $C_{\text{flow}}(s, T)$ is determined solely by the structural properties of R .

Justification. A representative sufficient instance is $h(s, T) = O(s + T)$, which models shared infrastructure—such as cloud orchestration platforms, automated scripting frameworks, or centralized delegation services—whose cost scales with the number of managed identities and the duration of deployment, but not with the product sT . This condition is consistent with empirically observed pooling behavior in deployed systems [7], [26], [27]. Examples include mining pools, validator delegation, and cloud-managed multi-account orchestration.

Role in the analysis. The structural results of Theorems 1 and 2 isolate resource-induced scaling independently of $h(s, T)$. Theorem 1 shows that even when $h(s, T) = 0$, a divisible, reusable resource may be insufficient to enforce linear identity cost within the present abstraction. Theorem 2 shows that a throughput-bounded resource enforces $C(s, T) = \Omega(sT)$ regardless of how small $h(s, T)$ is. The assumption therefore does not embed the conclusion: it captures the observation

that coordination overhead alone cannot restore linearity for parallelizable resources, nor can it eliminate the structural lower bound for throughput-bounded resources.

G. Identity-Amortization Property

Definition 9 (Identity-Amortizable Mechanism). *A mechanism is identity-amortizable if there exists a function $g(s, T)$ such that*

$$g(s, T) = o(sT) \quad \text{and} \quad C(s, T) \leq g(s, T)$$

for sufficiently large s .

Identity amortization implies

$$\lim_{s \rightarrow \infty} \frac{C(s, T)}{sT} = 0.$$

H. Structural Implication

The remainder of the paper establishes that divisible, reusable resources admit $C(s, T) = o(sT)$, while rate-limited resources enforce $C(s, T) = \Omega(sT)$. Adversarial identity scaling thus reduces to a classification problem over the resource R .

V. STRUCTURAL LIMITS OF PARALLELIZABLE RESOURCES

We establish that mechanisms grounded in divisible, reusable resources admit identity amortization within the considered model. The result is scoped to the present abstraction: it characterizes scaling behavior under the coordination model of Assumption 1.

A. Marginal Identity Cost

To study identity-level scaling, we define the marginal sustained identity cost

$$\Delta(s, T) = C(s, T) - C(s - 1, T).$$

A mechanism exhibits *vanishing marginal cost* if

$$\lim_{s \rightarrow \infty} \Delta(s, T) = 0.$$

Vanishing marginal cost implies identity amortization in a strong sense: each additional sustained identity becomes asymptotically free.

B. Adversarial Amortization Strategy

Let R be a parallelizable resource. By Definition 7, R satisfies divisibility, additivity of influence, temporal reusability, and identity transferability.

We construct an adversarial strategy that sustains s identities over T windows while keeping total resource expenditure independent of s , up to coordination overhead.

C. Identity Partition Invariance

Lemma 1. *Let R be divisible and additive. For any aggregate allocation $r \geq 0$ and any integer $s \geq 1$, the adversary can distribute r across s identities such that*

$$\sum_{v=1}^s f(r_v) = f(r) \quad \text{and} \quad \sum_{v=1}^s r_v = r.$$

Proof. By divisibility, r may be partitioned into allocations $r_1, \dots, r_s$ such that $\sum_{v=1}^s r_v = r$. By additivity,

$$\sum_{v=1}^s f(r_v) = f\left(\sum_{v=1}^s r_v\right) = f(r).$$

Thus influence is invariant to identity partitioning. $\square$

D. Cross-Window Reuse

Lemma 2. *If R is temporally reusable, a resource allocation r obtained in window t may be reused in all subsequent windows $t+1, \dots, T$ without additional expenditure.*

Proof. This follows directly from the definition of temporal reusability. $\square$

E. Coordination Overhead

The preceding arguments show that aggregate resource allocation can be reused across identities and time. However, maintaining multiple identities may incur additional coordination overhead. Let $h(s, T)$ denote the adversarial cost of coordinating s identities over T windows. The scaling behavior of $h(s, T)$ is governed by Assumption 1, stated in Section IV. The structural results are robust to any weaker condition $h(s, T) = o(sT)$, as noted therein.

F. Marginal Cost Collapse

Lemma 3. *Under the amortized coordination model, as $s, T \rightarrow \infty$ jointly,*

$$\frac{C(s, T)}{sT} \rightarrow 0,$$

and the marginal identity cost satisfies

$$\Delta(s, T) = O(1) = o(T).$$

Proof. By the activation threshold (Definition 1), sustaining s identities requires aggregate allocation at least $r \geq sr_{\min}$. For a parallelizable resource, the adversary acquires this aggregate *once*: by temporal reusability (Lemma 2), the allocation $r = sr_{\min}$ suffices for all T windows without renewed expenditure. By Lemma 1, r is then partitioned across s identities without changing total influence.

Thus the total adversarial cost satisfies

$$C(s, T) \leq sr_{\min} + h(s, T).$$

Under Assumption 1, $h(s, T) = o(sT)$. Therefore

$$\frac{C(s, T)}{sT} \leq \frac{r_{\min}}{T} + \frac{h(s, T)}{sT} \rightarrow 0 \quad \text{as } s, T \rightarrow \infty,$$

since the first term vanishes as $T \rightarrow \infty$ and the second by assumption. Hence $C(s, T) = o(sT)$.

For the marginal cost,

$$\Delta(s, T) \leq r_{\min} + [h(s, T) - h(s-1, T)] = O(1),$$

which is $o(T)$, confirming sublinear marginal cost. $\square$

G. Main Structural Limit Theorem

Theorem 1 (Structural Limit of Parallelizable Resources). *Let R be a parallelizable resource. Under the amortized coordination model of Assumption 1, any mechanism grounded in R is identity-amortizable within the present abstraction:*

$$C(s, T) = o(sT).$$

In particular, the marginal cost per additional identity satisfies $\Delta(s, T) = o(T)$: it does not scale with the time horizon.

Proof. By Lemma 3, under Assumption 1, as $s, T \rightarrow \infty$ jointly,

$$C(s, T) = o(sT).$$

Therefore any mechanism grounded in R is identity-amortizable by Definition 9. The marginal bound $\Delta(s, T) = o(T)$ follows directly from Lemma 3. $\square$

Remark 2. *Theorem 1 does not assert that a fixed aggregate resource can activate arbitrarily many identities: the activation threshold requires $r \geq sr_{\min}$, so aggregate resource must grow with s . The theorem asserts instead that a parallelizable resource imposes no additional per-window renewal cost per identity—once $r = sr_{\min}$ is acquired, temporal reusability eliminates the T -scaling overhead that throughput-bounded resources enforce. Moreover, protocol rules that merely redistribute or account for a fully parallelizable resource cannot by themselves create linear sustained identity cost; achieving $C(s, T) = \Omega(sT)$ requires introducing a structural constraint that breaks parallelizability.*

Thus, parallelizable resources enable identity amplification by allowing adversaries to acquire aggregate capacity once and redistribute it across arbitrarily many identities without per-window renewal overhead.

VI. LINEAR LOWER BOUND FOR THROUGHPUT-BOUNDED RESOURCES

We now establish the complementary structural result: throughput-bounded, non-transferable resources enforce linear sustained identity cost.

In contrast to parallelizable resources, where identities can be amortized across time and accounts, throughput-bounded resources require independent execution capacity per identity. This structural constraint prevents identity amplification and

forces adversarial cost to scale linearly with both the number of identities and the time horizon.

The result provides a formal foundation for mechanisms that achieve strong Sybil resistance through resource-induced constraints.

A. Structural Constraints

Let R be a throughput-bounded, non-transferable resource with window-local allocation.

Throughput boundedness implies the existence of a constant $\tau > 0$ such that for every identity v and window t

$$r_v(t) \leq \tau.$$

Thus a single identity cannot contribute more than τ units of resource per window.

We further assume that identity transferability fails: resource allocated to identity v cannot be reassigned to another identity without proportional cost.

Finally, resource allocation is window-local, meaning that allocation in window t does not automatically carry over to window $t + 1$. Sustained participation therefore requires renewed resource expenditure in every window.

Together, these constraints imply that each identity must be supported by an independent execution channel whose capacity cannot be shared or reused across identities or time.

B. Marginal Identity Cost

Recall the marginal sustained identity cost

$$\Delta(s, T) = C(s, T) - C(s - 1, T).$$

Lemma 4 (Positive Marginal Cost Under Throughput Constraints). *Let R be throughput-bounded, non-transferable, and window-local. Then there exists a constant $c > 0$ such that for all sufficiently large s*

$$\Delta(s, T) \geq cT.$$

Proof. To sustain s active identities, each identity must receive at least $r_{\min} > 0$ resource units per window by the activation threshold.

Thus sustaining s identities over T windows requires total allocation at least

$$sTr_{\min}.$$

Similarly sustaining $s - 1$ identities requires

$$(s - 1)Tr_{\min}.$$

Therefore

$$\Delta(s, T) = C(s, T) - C(s - 1, T) \geq Tr_{\min}.$$

Let $c = r_{\min}$. Then

$$\Delta(s, T) \geq cT.$$

□

C. Main Lower Bound Theorem

Theorem 2 (Linear Lower Bound). *Let R be throughput-bounded, non-transferable, and window-local. Then there exists a constant $c > 0$ such that*

$$C(s, T) \geq csT.$$

In particular

$$C(s, T) = \Omega(sT)$$

and the marginal sustained identity cost satisfies

$$\liminf_{s \rightarrow \infty} \Delta(s, T) \geq cT.$$

Proof. Fix s and T . Suppose the adversary sustains s active identities in each window $t = 1, \dots, T$. We show that no adversarial strategy can reduce total expenditure below $sTr_{\min}$, regardless of how identities are coordinated.

Step 1: Per-identity expenditure is unavoidable. By the activation threshold (Definition 1), each identity v must receive at least $r_{\min} > 0$ units of resource in every window t in which it is active. No protocol-level rule can substitute for this requirement, since activation is defined in terms of resource allocation, not identity labeling.

Step 2: Resource cannot be shared across identities. Since R is non-transferable (Definition 6, condition 2), the allocation $r_v(t)$ assigned to identity v in window t cannot be simultaneously credited to any other identity $v' \neq v$. Therefore the per-window expenditure satisfies

$$\sum_{v=1}^s r_v(t) \geq sr_{\min} \quad \text{for each } t = 1, \dots, T.$$

Step 3: Resource cannot be carried across windows. Since allocation is window-local (Definition 6, condition 3), the resource allocated in window t does not persist into window $t + 1$. The adversary must therefore renew the full per-identity expenditure independently in every window. Summing over all T windows,

$$C(s, T) \geq \sum_{t=1}^T sr_{\min} = sTr_{\min}.$$

Setting $c = r_{\min}$ completes the proof. The marginal lower bound $\liminf_{s \rightarrow \infty} \Delta(s, T) \geq cT$ follows directly from Lemma 4.

□

This establishes that throughput-bounded resources inherently limit adversarial scalability by eliminating the possibility of identity-level amortization. Each additional identity requires an independent execution channel, yielding linear cost growth under the constraints of Definition 6.

□

VII. EXTREMAL STRUCTURAL SEPARATION

We now combine the previous results to establish the extremal structural boundary in adversarial identity scaling.

Beyond a purely asymptotic distinction, this separation characterizes an asymptotic boundary between resource classes that enable identity amplification attacks and those that resist them under the considered model.

The result shows that adversarial scaling behavior is determined entirely by the structural properties of the underlying security-weighting resource.

A. Adversarial Scaling Geometry

Definition 10 (Scaling Geometry). *A mechanism induces*

- Sublinear scaling if $C(s, T) = o(sT)$,
- Linear scaling if $C(s, T) = \Omega(sT)$.

The scaling geometry characterizes how the adversarial cost of sustaining identities grows with the number of identities s and the time horizon T . It therefore determines whether identity amplification is feasible or structurally constrained.

B. Structural Separation Theorem

Theorem 3 (Extremal Structural Separation). *Let R be a security-weighting resource.*

- 1) *If R is parallelizable, the induced mechanism exhibits sublinear scaling.*
- 2) *If R is throughput-bounded, non-transferable, and window-local, the induced mechanism exhibits linear scaling.*

Consequently, parallelizable resources and throughput-bounded non-parallelizable resources induce disjoint adversarial scaling geometries.

Proof. Part (1) follows directly from the Structural Limit Theorem established in the previous section: parallelizable resources admit identity amortization, implying

$$C(s, T) = o(sT).$$

Part (2) follows from the Linear Lower Bound Theorem: throughput-bounded, non-transferable, window-local resources enforce

$$C(s, T) = \Omega(sT).$$

Since $o(sT)$ and $\Omega(sT)$ represent asymptotically distinct growth regimes, no mechanism can simultaneously exhibit both.

Therefore the two structural resource classes induce disjoint adversarial scaling geometries. $\square$

Thus, parallelizable resources enable identity amplification, whereas throughput-bounded resources fundamentally restrict it.

Corollary 1. *Within the structural model of this paper, linear sustained identity cost—and therefore resistance to identity amplification—cannot be enforced within the class of fully parallelizable resources under amortized coordination.*

C. Resource Substitution Principle

The separation result above yields a general design constraint for security mechanisms in open distributed systems.

Theorem 4 (Resource Substitution Principle). *Suppose a security mechanism aims to enforce linear sustained identity cost*

$$C(s, T) = \Omega(sT).$$

Then the underlying security-weighting resource cannot satisfy all structural properties of parallelizability.

In particular, at least one of the following properties must be violated:

- *Divisibility,*
- *Additivity of influence,*
- *Temporal reusability,*
- *Identity transferability.*

Proof. Assume for contradiction that the underlying resource satisfies all four properties above. By Definition 7, the resource is parallelizable.

From the Structural Limit Theorem, mechanisms grounded in parallelizable resources satisfy, within the present model,

$$C(s, T) = o(sT).$$

This contradicts the assumption that the mechanism enforces

$$C(s, T) = \Omega(sT).$$

Therefore any mechanism achieving linear sustained identity cost must violate at least one structural property defining parallelizability. $\square$

This principle can be interpreted as a design consideration: preventing identity amplification requires introducing structural limitations that break parallelizability.

The theorem highlights a fundamental design rule: linear identity-level Sybil resistance cannot be achieved through protocol design alone under the considered assumptions. Instead it requires selecting a resource class that structurally restricts parallelization, transferability, or temporal reuse.

D. Intermediate Structural Regimes

The separation established in Theorem 3 characterizes two extreme structural classes. However, real-world mechanisms may satisfy structural properties only partially, yielding scaling behavior that interpolates between the two asymptotic regimes established in this work.

Partial Transferability: Suppose identity transferability holds only up to a bounded fraction $\alpha \in [0, 1]$ of the aggregate allocation. That is, at most $\alpha \cdot r$ units of resource may be reassigned across identities, while the remaining $(1-\alpha) \cdot r$ units are identity-bound. In this case, the adversary can amortize only a fraction of the aggregate cost, yielding a mixed scaling regime. Specifically, one would expect:

$$C(s, T) = \Omega((1-\alpha) \cdot sT),$$

with the bound tightening as $\alpha \rightarrow 0$ (approaching full throughput-boundedness) and weakening as $\alpha \rightarrow 1$ (approaching full parallelizability).

Bounded Temporal Reuse: Suppose resource allocation persists across at most k consecutive windows rather than indefinitely. An adversary must then renew allocation every k windows, incurring cost proportional to $\lceil T/k \rceil$ renewal events. The resulting adversarial cost satisfies:

$$C(s, T) = \Omega\left(s \cdot \frac{T}{k}\right),$$

interpolating between linear scaling at $k = 1$ (window-local, full renewal) and sublinear scaling as $k \rightarrow T$ (full temporal reuse).

General Intermediate Regime: More generally, partial violations of parallelizability may yield cost functions satisfying

$$\omega(1) \leq C(s, T) \leq o(sT),$$

or other mixed asymptotic forms depending on which structural properties are partially relaxed and to what degree.

Characterizing these intermediate scaling geometries in full generality remains an open theoretical problem. A natural extension would reformulate the axiomatic framework using continuous structural parameters $(\alpha_{\text{div}}, \alpha_{\text{add}}, \alpha_{\text{reuse}}, \alpha_{\text{transfer}}) \in [0, 1]^4$ rather than binary property classes, and derive adversarial cost bounds as functions of these parameters.

The present work therefore establishes the extremal boundary conditions under which identity amortization is either unavoidable or asymptotically eliminated, leaving the interior of this parameter space as a direction for future work.

VIII. INSTANTIATIONS OF THROUGHPUT-BOUNDED RESOURCES

The throughput-bounded, non-transferable resource class introduced in Sections VI–VII is not merely a theoretical abstraction. These resources arise naturally in real-world systems and play a central role in preventing identity amplification attacks.

In contrast to parallelizable resources, which enable adversaries to scale identities at sublinear cost, throughput-bounded resources impose intrinsic per-identity constraints that eliminate amortization. Each additional identity must be supported by an independent execution channel, thereby enforcing linear adversarial cost.

This section characterizes concrete mechanisms that instantiate such constraints and illustrates how they fundamentally limit adversarial identity scaling. Figure 1 provides a conceptual overview of the relationship between resource structure and adversarial scaling geometry.

A. Per-Actor Throughput Channels

Let $\mathcal{U}$ denote a set of independent actors. Each actor $u \in \mathcal{U}$ is associated with a real-time execution channel $\mathcal{C}_u$ through which resource output is generated.

Definition 11 (Per-Actor Throughput Channel). *A channel $\mathcal{C}_u$ is throughput-bounded if there exists a constant $\tau > 0$ such that for every window t*

$$\text{rate}_t(\mathcal{C}_u) \leq \tau.$$

The quantity $\text{rate}_t(\mathcal{C}_u)$ represents the maximum resource output attributable to actor u during window t . Resource allocation for identity v is derived from the channel output:

$$r_v(t) \leq \text{rate}_t(\mathcal{C}_u).$$

This abstraction naturally induces throughput boundedness, identity non-transferability, and window-local temporal scope. Critically, it enforces that each identity must be backed by an independent execution channel, preventing cross-identity resource reuse.

B. Formal Classification

Theorem 5 (Per-Actor Channel Classification). *Let R be a resource derived from per-actor throughput channels. Then R satisfies (1) throughput boundedness, (2) failure of identity transferability, and (3) window-local temporal scope. Consequently R belongs to the throughput-bounded non-parallelizable class.*

Proof. Throughput boundedness follows from the existence of the per-channel rate limit τ . Identity transferability fails because resource generation is bound to the actor controlling channel $\mathcal{C}_u$. Window locality holds since each window requires fresh channel output. Thus the resource satisfies the defining properties of throughput-bounded non-parallelizable resources. $\square$

C. Examples of Throughput-Bounded Channels

Several real-world mechanisms naturally instantiate per-actor throughput channels.

a) Human Real-Time Participation: Human participation constitutes one concrete realization of a throughput-bounded channel. Human cognitive processing, attention span, and motor interaction impose intrinsic limits on sustained activity. These physiological constraints bound the amount of verifiable activity a single actor can produce per time window. Increasing computational infrastructure does not increase the throughput of a human-bound channel. Thus adversarial scaling requires recruiting additional independent actors, making identity amplification economically linear in s .

b) Device-Bound Execution: Hardware-bound environments can also induce throughput limits. Trusted execution environments, secure hardware tokens, and device attestation systems bind activity to physical hardware [58]. Each device can produce only limited verifiable output within a time window. An adversary cannot reuse a single device to sustain multiple identities within the same window, preventing cross-identity amortization.

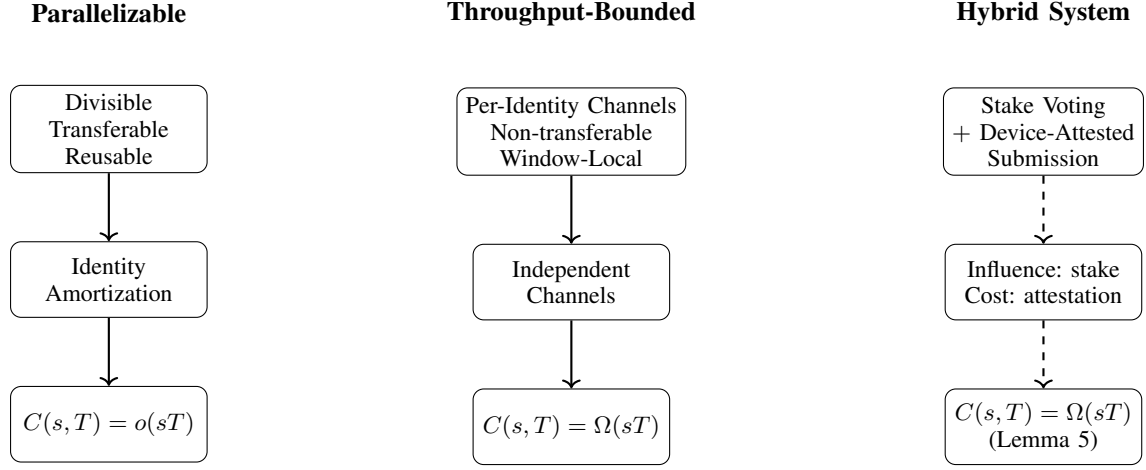


Fig. 1. Structural geometry of adversarial identity cost. Parallelizable resources (left) enable identity amortization and sublinear cost. Throughput-bounded resources (center) enforce independent per-identity channels and linear cost. A hybrid governance system (right) combines stake-weighted aggregate influence with device-attested proposal submission; by Lemma 5, under additive decomposition assumptions, the throughput-bounded component dominates and linear adversarial cost is preserved.

c) *Rate-Limited Network Participation:* Not all rate-limiting mechanisms instantiate throughput-bounded resources in the sense of this paper. Ordinary account-level rate limits do not qualify unless they are structurally bound to non-transferable actors, devices, or execution channels. If identities can be created cheaply and limits are attached only to syntactic account identifiers, adversaries may bypass such constraints through identity replication.

Throughput-boundedness therefore requires that rate constraints be tied to independent participation channels that cannot be aggregated or reassigned. Concrete examples include device-attested proposal submission, where each proposal must originate from a distinct attested hardware token, and hardware-rooted governance participation, where voting actions are bound to non-transferable execution environments. When such structural binding exists, per-identity execution constraints enforce adversarial cost that scales linearly with the number of identities.

D. Automation and AI

Automation does not invalidate the linear lower bound unless it eliminates the underlying throughput constraint. If participation requires independent channels whose throughput cannot be aggregated or transferred, scaling the number of identities still requires provisioning additional independent channels. The lower bound therefore depends on the persistence of actor-bound throughput constraints, not on assumptions about human uniqueness.

E. Hybrid Resource Models

Real-world systems rarely rely on a single security resource in isolation. Instead, deployed protocols often combine parallelizable and throughput-bounded resources. For example, a governance protocol may allocate voting power proportionally to financial stake while limiting proposal submission through per-identity rate constraints.

Lemma 5 (Hybrid Scaling Behavior). *Let $R = (R_p, R_t)$ where R_p is parallelizable and R_t is throughput-bounded and non-transferable. If adversarial cost decomposes as*

$$C(s, T) = C_p(s, T) + C_t(s, T),$$

then

$$C_p(s, T) = o(sT), \quad C_t(s, T) = \Omega(sT).$$

Consequently

$$C(s, T) = \Omega(sT).$$

Proof. We consider the adversary's cost minimization problem under the hybrid resource $R = (R_p, R_t)$.

Step 1: Additive decomposition assumption. We consider hybrid systems in which the total adversarial cost decomposes additively as

$$C(s, T) = C_p(s, T) + C_t(s, T),$$

representing independent resource constraints that contribute separately to adversarial expenditure.

Step 2: Bounding each component. By Theorem 1, $C_p(s, T) = o(sT)$ since R_p is parallelizable and admits identity amortization. By Theorem 2, $C_t(s, T) \geq c \cdot sT$ for some constant $c > 0$, since R_t is throughput-bounded, non-transferable, and window-local.

Step 3: The throughput-bounded component dominates. Since $C_t(s, T) = \Omega(sT)$ and $C_p(s, T) = o(sT)$, for sufficiently large s the throughput-bounded term dominates:

$$C(s, T) = C_p(s, T) + C_t(s, T) \geq C_t(s, T) \geq c \cdot sT.$$

Therefore $C(s, T) = \Omega(sT)$. □

This result applies to additive decompositions of hybrid resources and should be interpreted as a sufficient structural characterization, rather than a complete description of

all hybrid compositions. Non-additive hybrid compositions, in which the two resource components interact or partially substitute for one another, are outside the scope of this paper and remain an open direction for future work.

Thus, throughput-bounded components act as a bottleneck that neutralizes identity amplification attacks enabled by parallelizable resources.

F. Analytical Illustration of the Scaling Laws

To illustrate the asymptotic bounds proved in Theorems 1–3 we consider closed-form expressions derived directly from the theoretical analysis. These curves are illustrative rather than empirical; they visualize the theoretical scaling behavior established in Theorems 1 and 2.

For parallelizable resources the adversary incurs a fixed aggregate allocation cost

$$C_p(s, T) = r_{\text{fixed}},$$

while throughput-bounded resources require per-identity resource expenditure

$$C_t(s, T) = r_{\text{min}} \cdot s \cdot T.$$

Figure 2 plots these expressions for $s \in [1, 10,000]$ with $T = 100$ windows. The parallelizable cost curve is flat, reflecting identity amortization and vanishing marginal identity cost. In contrast, the throughput-bounded curve grows linearly, confirming the asymptotic lower bound $C(s, T) = \Omega(sT)$.

G. Practical Enforcement of Throughput Bounds

The theoretical results developed in this paper assume the existence of non-transferable per-identity throughput bounds. In practice such constraints are approximated through mechanisms that bind resource generation to independent actors, devices, or execution environments.

Examples include rate-limited protocol participation, device-attested computation, and human-interaction tasks. While no practical mechanism perfectly enforces strict non-transferability, systems that approximate these structural properties inherit the linear adversarial cost guarantees established in this work and effectively prevent identity amplification.

IX. ADVERSARIAL IMPLICATIONS AND ATTACK SURFACES

The asymptotic separation of Sections V–VII has direct adversarial implications. When a resource is divisible and reusable, adversarial influence may be redistributed across identities while preserving aggregate capacity, enabling attacks that remain invisible under aggregate-weight security models. The key shift is from *how much resource an adversary controls* to *how adversarial cost scales as identities proliferate*.

A. Canonical Amortization Attack

Let R be a divisible, reusable resource. An adversary: (i) acquires aggregate allocation $r = sr_{\text{min}}$; (ii) partitions r across s identities (divisibility); (iii) reassigns allocations via transferability; and (iv) reuses allocations across windows (temporal reusability). By Lemmas 1–3, total cost satisfies

$$C(s, T) \leq sr_{\text{min}} + h(s, T) = o(sT),$$

so marginal cost $\Delta(s, T) = O(1) = o(T)$. The adversary expands the identity set without per-window renewal overhead—in contrast to rate-limited resources, where $\Delta(s, T) = \Omega(T)$.

B. Instantiation in Blockchain Systems

a) Proof-of-Work. PoW is a hybrid. Mining capacity (hardware) is divisible and transferable; mining pools implement identity partitioning by distributing hash power while maintaining aggregate control. Theorem 1 applies to this capacity component. *Operational expenditure* (energy) is non-reusable and window-local, contributing a flow cost not captured by the divisible classification. The full economic cost of sustaining multiple mining identities combines both components.

b) Proof-of-Stake. Financial stake satisfies all four divisibility properties. Delegation markets enable a fixed capital base to be distributed across arbitrarily many validators. Under the considered model, stake therefore admits identity amortization: $C(s, T) = o(sT)$ with respect to the stake component.

C. Case Study: Stake-Governed Protocol

Consider a governance protocol where voting influence is proportional to delegated stake, proposal submission is rate-limited to one per identity per epoch, and committee membership is determined by per-identity sortition.

Attack. An adversary controlling 1% of total stake S splits across $s = 100$ accounts, each holding $0.01S/100$. Total voting weight remains $0.01S$; proposal bandwidth expands to 100 proposals per epoch. Since stake redistribution incurs no additional T -scaling expenditure, this amplification is essentially free under the considered model.

Structural analysis. The stake component admits amortization by Theorem 1. If the rate limit is bound to non-transferable execution channels (e.g., device attestation), then by Theorem 2 the participation component enforces $C_t(s, T) = \Omega(sT)$, and by Lemma 5 total cost satisfies $C(s, T) = \Omega(sT)$. If the rate limit is enforced only at the account level, without structural binding, the constraint is bypassable and the system reduces to the divisible regime of Theorem 1.

Design implication. Identity-level security in hybrid protocols depends not on rate limits alone, but on whether those limits are structurally bound to non-transferable participation channels.

D. Governance Attack Surface

In identity-sensitive governance, an adversary controlling total stake R may distribute it across s identities without

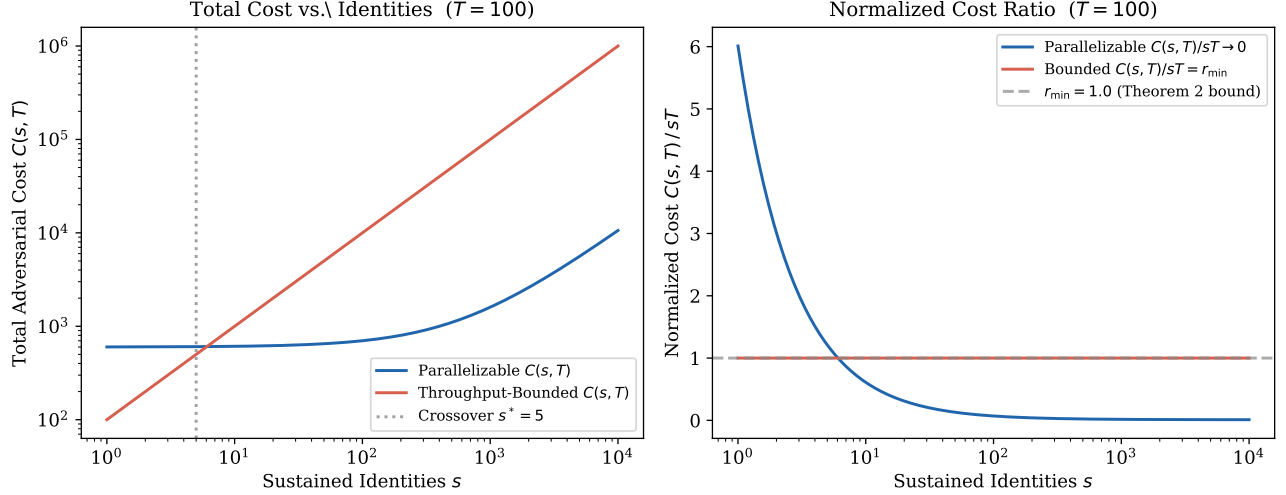


Fig. 2. Adversarial cost geometry derived from Theorems 1–3. *Left*: total adversarial cost $C(s, T)$ vs. sustained identities s ; parallelizable resources exhibit cost growing only with s (stock acquisition) whereas throughput-bounded resources enforce linear growth in both s and T . The crossover $s^* = r_{\text{fixed}}/(r_{\min}T)$ marks where throughput-bounded cost exceeds the parallelizable baseline. *Right*: normalized total cost $C(s, T)/sT$; the parallelizable ratio declines as $s, T \rightarrow \infty$ jointly (Theorem 1), while the throughput-bounded ratio remains bounded away from zero at $r_{\min}$ (Theorem 2). Note that $C(s, T)/sT$ measures amortized cost per identity per window and should not be confused with marginal identity cost $\Delta(s, T) = C(s, T) - C(s - 1, T)$, which for throughput-bounded resources satisfies $\Delta(s, T) \geq r_{\min}T$.

changing total voting weight, yet gain: increased proposal capacity, biased committee selection, amplified visibility in deliberation, and influence over rate-limited participation. These effects arise without increasing aggregate resource ownership and are not captured by classical aggregate-weight analyses.

E. Mitigation and Design Principle

Under the considered model, introducing per-identity throughput constraints bound to non-transferable channels may restore $C(s, T) = \Omega(sT)$. This suggests treating identity-level security as a co-design problem between protocol rules and resource structure. Systems that rely exclusively on divisible, reusable resources may admit amortization properties under the considered model; incorporating at least one rate-limited, non-transferable component appears necessary to resist identity amplification within this framework.

Examples of such components include: actor-bound participation constraints, device-attested execution, non-transferable throughput tokens, or interaction channels requiring independent execution.

X. LIMITATIONS AND OPEN PROBLEMS

While the structural separation established in this work identifies asymptotic constraints on adversarial identity scaling within the present model, several limitations and open questions remain.

A. Measurement of Throughput Constraints

Our analysis assumes the existence of a per-identity throughput bound τ and an activation threshold $r_{\min}$. In practice, accurately measuring and enforcing such constraints may be challenging.

Real-world systems must ensure that throughput limits cannot be bypassed through aggregation, outsourcing, or hardware replication. If the effective per-identity throughput increases with additional infrastructure or automation, the linear lower bound may weaken.

Designing robust mechanisms that enforce non-transferable, rate-limited participation therefore remains an important engineering challenge.

B. Verification and Measurement Overhead

Even when throughput-bounded resources exist, systems must verify that resource usage actually corresponds to independent identities or actors.

Verification mechanisms—such as device attestation, interaction verification, or behavioral validation—may introduce additional complexity, latency, or economic cost. The present model excludes verification cost from $C(s, T)$; in practice, such costs may be non-negligible and should be incorporated into any concrete deployment analysis.

Understanding the trade-offs between strong throughput verification and system scalability remains an open systems-design problem.

C. Coordinated Multi-Actor Adversaries

The lower bound shows that sustaining s identities requires $\Omega(sT)$ resource expenditure when each identity must be supported by an independent throughput channel.

However, adversaries may coordinate large numbers of actors. While such coordination does not violate the linear bound, it raises additional questions regarding economic feasibility, incentive compatibility, and the potential emergence of identity markets or participation outsourcing.

Analyzing equilibrium behavior under large-scale coordinated participation remains an important direction for future work.

D. Hybrid Resource Models

Many real-world systems combine multiple resource classes, such as capital, computational effort, and human participation.

Our separation theorem applies cleanly to pure resource classes, but hybrid constructions may exhibit intermediate scaling behavior. For example, systems that combine stake-weighted influence with rate-limited participation may simultaneously exhibit parallelizable and throughput-bounded properties.

Partial results for additive hybrid compositions are established in Lemma 5 and Section VIII-E, where the throughput-bounded component is shown to dominate asymptotically under additive cost decomposition. Characterizing adversarial scaling laws for non-additive hybrid compositions and more general resource interactions remains an open theoretical problem.

E. Adaptive and Evolving Resource Classes

Technological evolution may alter the structural properties of resources over time. For example, advances in automation or AI-driven participation may increase effective throughput or reduce non-transferability.

Our framework provides a structural lens for analyzing such changes, but the validity of resource assumptions must be periodically reassessed as system environments evolve.

F. Beyond Linear Scaling

This work distinguishes between sublinear and linear identity-scaling regimes. Other scaling geometries—such as superlinear or convex adversarial cost growth—may arise under stronger structural constraints.

Identifying minimal structural conditions that enforce stronger-than-linear identity cost remains an open theoretical direction.

G. Protocol-Level Realization

The results of this paper are structural and protocol-agnostic. Translating these guarantees into deployable systems requires careful engineering of verification mechanisms, incentive compatibility, and liveness properties.

Bridging the structural theory of security resources with fully composable distributed implementations remains a promising avenue for future research.

H. Adaptive Economic Incentives and Governance Dynamics

The present framework models adversarial cost as a deterministic function of resource allocation, abstracting away adaptive economic behavior, endogenous governance dynamics, and evolving participation incentives. In practice, sustained identity cost may be influenced by market prices for delegated

resources, token inflation, regulatory changes, and governance rules that adjust thresholds in response to observed participation patterns. The framework intentionally isolates structural resource properties rather than modeling complete governance ecosystems. Incorporating adaptive incentive structures and endogenous governance dynamics remains an important direction for future work.

I. Automation and Evolving Throughput

The linear lower bound for rate-limited resources rests on the assumption that per-identity throughput constraints are enforced and cannot be circumvented by additional infrastructure. Advances in automation, AI-driven participation, and hardware acceleration may progressively reduce the effective cost of provisioning independent execution channels, potentially narrowing the practical gap between the two scaling regimes even when the theoretical separation holds asymptotically. The framework provides a structural lens for analyzing such changes, but the validity of throughput assumptions should be periodically reassessed as deployment environments evolve.

J. Socio-Economic Behaviors and Identity Markets

The model treats adversarial identity creation as a direct resource acquisition problem, abstracting away socio-economic coordination mechanisms. In practice, identity markets, participation outsourcing, and secondary markets for rate-limited credentials may alter the effective cost structure in ways not captured by the abstraction, potentially weakening non-transferability assumptions in practice even when they hold structurally. Analyzing equilibrium behavior in such settings remains outside the scope of the present work.

K. Scope of the Framework

The results developed in this paper characterize adversarial identity scaling under an abstract structural model. The framework is not intended as a complete empirical taxonomy of deployed systems, nor does it claim that all real-world systems can be perfectly classified into purely divisible or rate-limited categories. Rather, it isolates structural properties that govern asymptotic scaling behavior under explicit assumptions, providing a complementary perspective alongside deployment-specific and protocol-level analyses. Real deployments may exhibit partial transferability, imperfect throughput enforcement, or hybrid resource composition; in such settings, the scaling geometry may interpolate between the regimes characterized here.

L. Window Size and Temporal Granularity

The system model partitions time into discrete windows of uniform size, and the adversarial cost function $C(s, T)$ is defined over a horizon of T such windows. The choice of window size is therefore a modeling parameter that affects how the results apply to concrete systems.

Smaller windows correspond to finer temporal granularity, increasing the number of renewal events required for sustained

participation under throughput-bounded resources. In the limit of very small windows, the per-window activation cost $r_{\min}$ may approach zero while T grows proportionally, leaving the product $r_{\min} \cdot T$ roughly constant. The linear lower bound $C(s, T) \geq r_{\min} \cdot sT$ therefore remains meaningful provided that $r_{\min}$ and T are calibrated consistently with the chosen window granularity.

Larger windows reduce the number of renewal events but may allow adversaries to concentrate resource expenditure within a single window, potentially relaxing the window-locality constraint. Systems in which resource allocation persists across multiple windows should be modeled with correspondingly larger $r_{\min}$ or reduced effective T .

More generally, the window size assumption interacts with the throughput bound τ and the activation threshold $r_{\min}$ in ways that depend on the deployment context. Characterizing the sensitivity of the scaling bounds to window granularity—and identifying window-size-independent formulations of the structural separation—remains an open direction for future work.

XI. ANALYTICAL ILLUSTRATIONS OF SCALING REGIMES

We visualize the cost functions implied by Theorems 1 and 2 under parametric variation. These are analytical illustrations of asymptotic scaling behavior derived from the theoretical model—not empirical benchmarks or experimental proofs.

A. Setup

We evaluate the model cost functions directly: $C_{\text{par}}(s, T) = s \cdot r_{\min} + h(s, T)$ for divisible resources (one-time stock acquisition plus coordination overhead), and $C_{\text{bnd}}(s, T) = s \cdot T \cdot r_{\min}$ for rate-limited resources (mandatory per-identity, per-window expenditure). Throughout, $r_{\min} = 1.0$. For E4, three coordination models are tested: $h_1 = s + T$, $h_2 = s \log T$, $h_3 = s\sqrt{T}$, all satisfying $h(s, T) = o(sT)$.

B. E1: Identity Scaling ($T = 100$, vary s)

The normalized cost $C_{\text{par}}(s, T)/sT$ decreases monotonically from 1.02 at $s = 1$ to 0.03 at $s = 100$, consistent with $C(s, T) = o(sT)$ (Theorem 1). The rate-limited ratio remains at 1.0 for all s , consistent with $C(s, T) = \Theta(sT)$ (Theorem 2).

TABLE IV
E1: NORMALIZED ADVERSARIAL COST $C(s, T)/sT$ ($T = 100$, $r_{\min} = 1.0$, $h = s + T$).

s	C_{par}	C_{par}/sT	C_{bnd}	C_{bnd}/sT
1	102.0	1.0200	100.0	1.0000
5	110.0	0.2200	500.0	1.0000
10	120.0	0.1200	1000.0	1.0000
25	150.0	0.0600	2500.0	1.0000
50	200.0	0.0400	5000.0	1.0000
100	300.0	0.0300	10000.0	1.0000

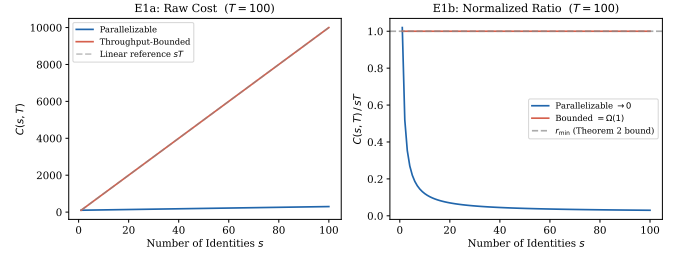


Fig. 3. E1: Normalized adversarial cost $C(s, T)/sT$ vs. s ($T = 100$). Divisible-resource ratio decays toward zero (Theorem 1); rate-limited ratio is constant at $r_{\min}$ (Theorem 2).

C. E2: Time Horizon Scaling ($s = 10$, vary T)

C_{par} grows as $O(s + T)$, well below sT , while C_{bnd} tracks $sTr_{\min}$ exactly. The normalized divisible ratio continues to decrease as T grows, consistent with Theorem 1.

TABLE V
E2: NORMALIZED COST $C(s, T)/sT$ ($s = 10$, $r_{\min} = 1.0$, $h = s + T$).

T	C_{par}	C_{par}/sT	C_{bnd}	C_{bnd}/sT
10	30.0	0.3000	100.0	1.0000
25	45.0	0.1800	250.0	1.0000
50	70.0	0.1400	500.0	1.0000
100	120.0	0.1200	1000.0	1.0000
150	170.0	0.1133	1500.0	1.0000
200	220.0	0.1100	2000.0	1.0000

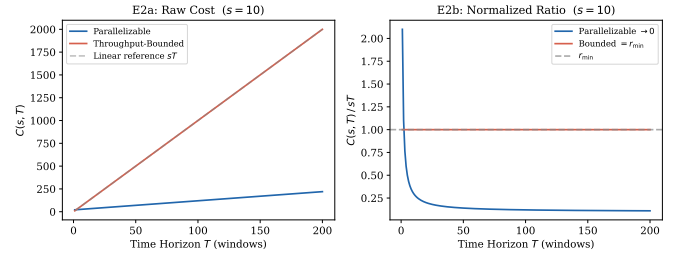


Fig. 4. E2: $C(s, T)$ vs. T ($s = 10$). Divisible cost grows sublinearly; rate-limited cost is linear in T (Theorems 1–2).

D. E3: Marginal Identity Cost ($T = 100$)

$\Delta_{\text{par}}(s, T) = O(1)$ for all s ; $\Delta_{\text{bnd}}(s, T) = T \cdot r_{\min} = 100$ throughout. The latter is consistent with the lower bound $\liminf_{s \rightarrow \infty} \Delta(s, T) \geq r_{\min}T$ established in Theorem 2.

E. E4: Coordination Overhead Robustness ($T = 100$)

All three models yield $C_{\text{par}}(s, T)/sT \rightarrow 0$, well below the rate-limited baseline of 1.0. The sublinear regime is robust across the class $h(s, T) = o(sT)$, consistent with Assumption 1.

Illustrations E1–E4 are drawn from the theoretical cost functions and do not constitute experimental evidence about deployed systems. They confirm that both scaling regimes are

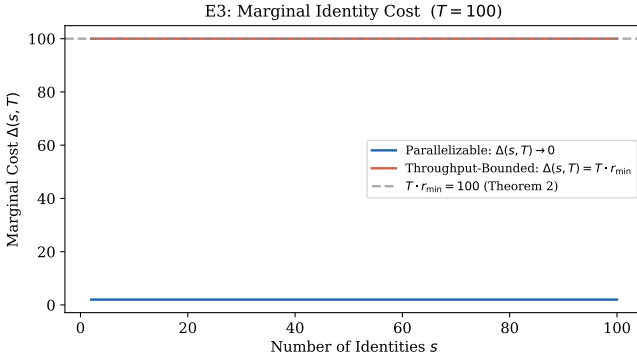


Fig. 5. E3: Marginal cost $\Delta(s, T)$ vs. s ($T = 100$). Rate-limited: $\Delta = T \cdot r_{\min} = 100$ (Theorem 2 lower bound, dashed). Divisible: $\Delta = O(1) = o(T)$ (Theorem 1).

TABLE VI

E4: $C_{\text{par}}(s, T)/sT$ UNDER THREE OVERHEAD MODELS ($T = 100$, $r_{\min} = 1.0$).

s	$h_1=s+T$	$h_2=s \log T$	$h_3=s\sqrt{T}$	Bounded
1	1.0200	0.0562	0.1100	1.0000
5	0.2200	0.0562	0.1100	1.0000
10	0.1200	0.0562	0.1100	1.0000
25	0.0600	0.0562	0.1100	1.0000
50	0.0400	0.0562	0.1100	1.0000
100	0.0300	0.0562	0.1100	1.0000

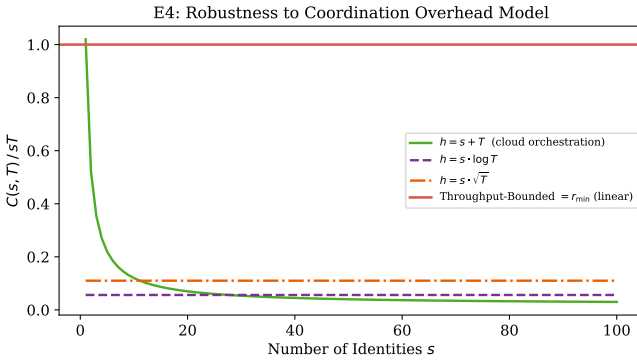


Fig. 6. E4: $C_{\text{par}}(s, T)/sT$ under three overhead models, all satisfying $h = o(sT)$. All decay toward zero (Theorem 1); rate-limited baseline (solid) remains at $r_{\min}$ (Theorem 2).

consistent across the model parameter space, and that the marginal cost bound of Theorem 2 is tight.

XII. CONCLUSION

This paper proposed an axiomatic framework for reasoning about adversarial identity scaling in open systems through the lens of resource structure. The central contribution is not a universally complete characterization of Sybil resistance, but a resource-centric asymptotic perspective that complements existing protocol-level and economic analyses.

We formalized resource properties—divisibility, additivity of influence, temporal reusability, identity transferability, and

throughput boundedness—and showed that these properties indicate the asymptotic geometry of sustained adversarial cost under the coordination model considered. Within the present abstraction, resources satisfying divisibility and reusability admit identity amortization, suggesting a structural tendency toward sublinear per-identity cost. Conversely, rate-limited, non-transferable, window-local resources indicate the scaling regime $C(s, T) = \Omega(sT)$.

Analytical illustrations confirmed that these two scaling regimes are consistent across varying identity counts, time horizons, and coordination overhead models within the framework, and that the marginal cost bound associated with rate-limited resources is tight. These illustrations are drawn from the theoretical cost functions and do not constitute experimental evidence about deployed systems.

Taken together, these results suggest an asymptotic boundary in the design space of open systems. Under the assumptions considered, mechanisms grounded in fully divisible and reusable resources may be insufficient to enforce linear per-identity cost under the considered assumptions. Security guarantees at the identity layer are therefore constrained not only by aggregate resource scarcity, but by the structural properties of the resource used to regulate participation.

This perspective is intended as a complement to, rather than a replacement of, existing security analyses. Classical consensus models and economic equilibrium analyses capture important adversarial dimensions outside the scope of this work. The framework here isolates a different dimension: the degree to which resource divisibility and reusability enable adversarial cost amortization across identities and time.

Future work includes characterization of hybrid resource compositions, mechanisms for enforcing practical throughput bounds, analysis of equilibrium behavior under adaptive economic incentives, and empirical measurement of identity-scaling behavior in live open systems.

APPENDIX A

TIGHTNESS AND ROBUSTNESS OF THE SCALING BOUNDS

This appendix establishes that the asymptotic bounds derived in Theorems 1 and 2 are tight and remain robust under several extensions of the adversarial model.

In particular, we construct explicit adversarial strategies that match the asymptotic upper and lower bounds associated with the two structural resource classes introduced in this work. These constructions demonstrate that the separation between sublinear and linear identity-scaling regimes is not merely qualitative, but asymptotically exact.

A. Tightness for Parallelizable Resources

Let R be a parallelizable resource satisfying divisibility, additivity of influence, temporal reusability, and identity transferability.

By the activation threshold (Definition 1), sustaining s identities requires aggregate allocation at least $r \geq s r_{\min}$. For a parallelizable resource the adversary acquires this aggregate *once*: by temporal reusability, the allocation $r = s r_{\min}$ suffices for all T windows without renewed expenditure.

By divisibility, the adversary partitions r into components $r_1, \dots, r_s$ with $\sum_{i=1}^s r_i = r$. By additivity of influence,

$$\sum_{i=1}^s f(r_i) = f\left(\sum_{i=1}^s r_i\right) = f(r).$$

Hence total influence is invariant under identity partitioning. The total adversarial cost satisfies

$$C(s, T) \leq s r_{\min} + h(s, T).$$

Under Assumption 1, $h(s, T) = o(sT)$. Therefore

$$\frac{C(s, T)}{sT} \leq \frac{r_{\min}}{T} + \frac{h(s, T)}{sT} \rightarrow 0 \quad \text{as } s, T \rightarrow \infty \text{ jointly,}$$

since the first term vanishes as $T \rightarrow \infty$ and the second by assumption.

Thus $C(s, T) = o(sT)$, confirming that the sublinear scaling regime of Theorem 1 is asymptotically tight.

B. Tightness for Throughput-Bounded Resources

Let R be a throughput-bounded resource satisfying Definition 6, with activation threshold $r_{\min}$ and window-local allocation.

Consider an adversary sustaining s identities over T windows.

By construction, each identity must receive at least $r_{\min}$ units of resource within every active window. Allocating

$$r_v(t) = r_{\min}$$

for all identities v and windows t yields total cost

$$C(s, T) = \sum_{t=1}^T \sum_{v=1}^s r_{\min}.$$

Evaluating the sum gives

$$C(s, T) = sTr_{\min}.$$

Therefore,

$$C(s, T) = \Theta(sT),$$

which matches the lower bound established in Theorem 2.

Hence the linear scaling regime is also asymptotically tight.

Together, these constructions establish that the two asymptotic regimes identified in this paper represent tight structural characterizations of adversarial identity scaling.

APPENDIX B

ROBUSTNESS UNDER EXTENDED ADVERSARIAL MODELS

We now examine the robustness of the scaling results under natural extensions of the adversarial model.

The objective is not to characterize every possible adversarial behavior, but to demonstrate that the structural separation remains stable under several realistic variations.

A. Adaptive Identity Creation

Suppose the adversary creates identities adaptively over time.

Under parallelizable resources, adaptive creation does not alter the scaling regime, since aggregate resource ownership may still be repartitioned across newly created identities.

Under throughput-bounded resources, each newly activated identity must continue to satisfy the per-window activation threshold. Therefore adaptive creation preserves the lower bound

$$C(s, T) = \Omega(sT).$$

B. Coalitions of Independent Actors

Suppose an adversary coordinates k independent actors, each associated with a throughput-bounded execution channel.

Although aggregate throughput increases with k , sustaining s concurrent identities still requires at least s independent channels.

Thus coalition formation does not invalidate the linear lower bound, and

$$C(s, T) = \Omega(sT)$$

continues to hold.

C. Stochastic Throughput Bounds

Consider a setting in which throughput is stochastic but bounded in expectation.

Let $\tau_v(t)$ denote the random throughput available to identity v during window t , with bounded expectation

$$\mathbb{E}[\tau_v(t)] \leq \tau.$$

If sustained participation requires expected activation threshold $r_{\min}$ per identity per window, then expected total expenditure satisfies

$$\mathbb{E}[C(s, T)] = \Omega(sT).$$

Under standard concentration assumptions, deviations from the expected cost remain bounded with high probability, preserving linear scaling asymptotically.

D. Partial Relaxation of Structural Properties

The framework developed in this paper characterizes two extremal structural regimes: fully parallelizable resources and fully throughput-bounded resources.

Intermediate systems may partially satisfy transferability, temporal locality, or throughput boundedness. In such cases, adversarial scaling behavior may interpolate between the asymptotic regimes established in Theorems 1–2.

For example, if only a fraction of aggregate allocation is transferable across identities, the adversary may amortize only that component of the resource. Likewise, bounded temporal reuse across a limited number of windows may reduce, but not eliminate, sustained expenditure.

These intermediate cases suggest a continuum of scaling behaviors between sublinear amortization and linear growth.

Formal characterization of such transition regimes, including potential phase boundaries between amortizable and non-amortizable resources, remains an important direction for future work.

REFERENCES

- [1] J. R. Douceur, “The sybil attack,” in *IPTPS*, 2002, pp. 251–260.
- [2] Lamport *et al.*, “Byzantine generals,” *TOPLAS*, 1982.
- [3] Fischer *et al.*, “Flp impossibility,” *JACM*, 1985.
- [4] Castro and Liskov, “Pbft,” in *OSDI*, 1999.
- [5] J. Garay, A. Kiayias, and N. Leonardos, “The bitcoin backbone protocol,” in *EUROCRYPT*, 2015, pp. 281–310.
- [6] L. Ren, “Analysis of nakamoto consensus,” in *EUROCRYPT*, 2019.
- [7] Bonneau *et al.*, “Sok bitcoin,” in *IEEE S&P*, 2015.
- [8] Narayanan *et al.*, *Bitcoin Book*. Princeton, 2016.
- [9] Gencer *et al.*, “Decentralization,” in *FC*, 2018.
- [10] Decker and Wattenhofer, “Bitcoin propagation,” in *IEEE P2P*, 2013.
- [11] A. Kiayias, A. Russell, B. David, and R. Oliynykov, “Ouroboros,” in *CRYPTO*, 2017.
- [12] Dwork *et al.*, “Partial synchrony,” *JACM*, 1988.
- [13] Gervais *et al.*, “Pow security,” in *ACM CCS*, 2016.
- [14] Eyal and Sirer, “Selfish mining,” in *FC*, 2014.
- [15] Carlsten *et al.*, “Bitcoin instability,” in *ACM CCS*, 2016.
- [16] Sapirshtein *et al.*, “Optimal selfish mining,” in *FC*, 2016.
- [17] Eyal, “Miner’s dilemma,” in *IEEE S&P*, 2015.
- [18] Apostolaki *et al.*, “Routing attacks,” in *IEEE S&P*, 2017.
- [19] Neudecker *et al.*, “Network attacks,” in *IEEE S&P*, 2019.
- [20] A. Gencer *et al.*, “Scaling nakamoto consensus to thousands of transactions per second,” in *Financial Cryptography*, 2018.
- [21] B. Biais *et al.*, “Blockchain consensus protocols in the wild,” in *Review of Financial Studies*, 2019.
- [22] V. Buterin, “Daos, dacs, das and more: An incomplete terminology guide,” <https://blog.ethereum.org/2014/05/06/daos-dacs-das-and-more-an-incomplete-terminology-guide>, 2014.
- [23] Gilad *et al.*, “Algorand,” in *SOSP*, 2017.
- [24] P. Resnick *et al.*, “Reputation systems,” *Communications of the ACM*, 2000.
- [25] J. Li *et al.*, “Sybilcontrol: Practical sybil defense with computational puzzles,” in *IEEE INFOCOM*, 2017.
- [26] M. Rosenfeld, “Analysis of bitcoin pooled mining reward systems,” *arXiv preprint arXiv:1112.4980*, 2011.
- [27] F. Saleh, “Blockchain without waste: Proof-of-stake,” *Review of Financial Studies*, 2021.
- [28] Bano *et al.*, “Consensus in blockchains,” in *ACM AFT*, 2019.
- [29] E. Budish, “Economic limits of blockchain,” *NBER*, 2018.
- [30] Budish *et al.*, “Economic limits of consensus,” in *ACM EC*, 2024.
- [31] Miller *et al.*, “Honeybadgerbft,” in *ACM CCS*, 2016.
- [32] Sompolinsky and Zohar, “Ghost,” in *FC*, 2015.
- [33] H. Yu *et al.*, “Sybilguard,” in *SIGCOMM*, 2006.
- [34] —, “Sybillimit,” in *IEEE S&P*, 2008.
- [35] G. Danezis and P. Mittal, “Sybilinfer,” in *NDSS*, 2009.
- [36] S. Wang *et al.*, “Attack of the clones: Sybil attacks in blockchain,” in *ACSAC*, 2019.
- [37] D. Laszka *et al.*, “A game-theoretic analysis of sybil attacks,” in *AAMAS*, 2020.
- [38] R. Pass and E. Shi, “Hybrid consensus,” in *DISC*, 2017.
- [39] —, “Fruitchains,” in *PODC*, 2017.
- [40] I. Eyal *et al.*, “Bitcoin-ng: A scalable blockchain protocol,” in *NSDI*, 2016.
- [41] Kroll *et al.*, “Bitcoin economics,” in *WEIS*, 2013.
- [42] M. Rosenfeld, “Analysis of bitcoin pooled mining reward systems,” in *arXiv*, 2014.
- [43] F. Tramèr *et al.*, “The slush pool attack,” in *USENIX Security*, 2020.
- [44] M. Borge *et al.*, “Proof-of-personhood,” in *IEEE EuroS&P Workshops*, 2017.
- [45] Ford, “Pseudonym parties,” in *IEEE S&P Workshops*, 2020.
- [46] S. Nakamoto, “Bitcoin: A peer-to-peer electronic cash system,” <https://bitcoin.org/bitcoin.pdf>, 2008.
- [47] A. Kiayias *et al.*, “Ouroboros praos,” in *EUROCRYPT*, 2018.
- [48] —, “Ouroboros genesis,” in *ACM CCS*, 2020.
- [49] Y. Gilad *et al.*, “Algorand extended analysis,” in *Cryptology ePrint*, 2017.
- [50] A. Gervais *et al.*, “Is bitcoin a decentralized currency?” in *IEEE Security & Privacy*, 2014.
- [51] J. Alwen *et al.*, “Blockchain protocols in the wild,” in *CRYPTO*, 2019.
- [52] Heilman *et al.*, “Eclipse attacks,” in *USENIX Security*, 2015.
- [53] S. Micali *et al.*, “Verifiable random functions,” in *FOCS*, 1999.
- [54] E. Syta *et al.*, “Scalable bias-resistant distributed randomness,” in *IEEE S&P*, 2017.
- [55] R. Pass and E. Shi, “The sleepy model,” in *ASIACRYPT*, 2017.
- [56] Levy, *Capability Systems*. Digital Press, 1984.
- [57] Jacobson, “Congestion control,” *SIGCOMM*, 1988.
- [58] Costan and Devadas, “Intel sgx,” in *IACR*, 2016.